Web Development Foundations

Week 8 – "Async Time Matters"

Day 1: "Async – Time Matters"

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Async Is About Time

Async is about time, *not magic*.

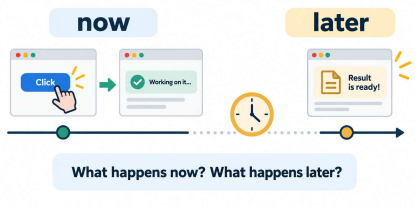
Some work:

- happens now
- starts now but finishes later
- needs the page to stay understandable while waiting

Today we ask:

- What happens now?
- What happens later?

2



Async Is About Time

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Lab Review: Successful Path

Previous lab review:

- working behavior
- named functions
- clear responsibilities
- same result after refactoring

Success path:

Structure helps us see where timing belongs.



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From Structure To Timing

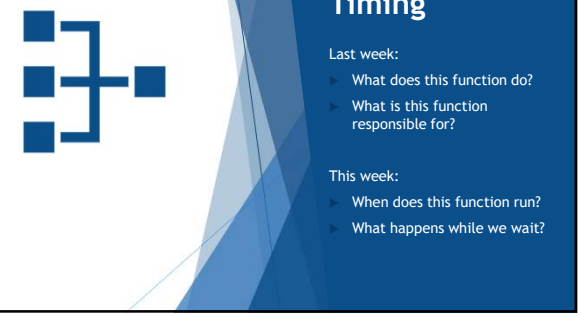
Last week:

- What does this function do?
- What is this function responsible for?

This week:

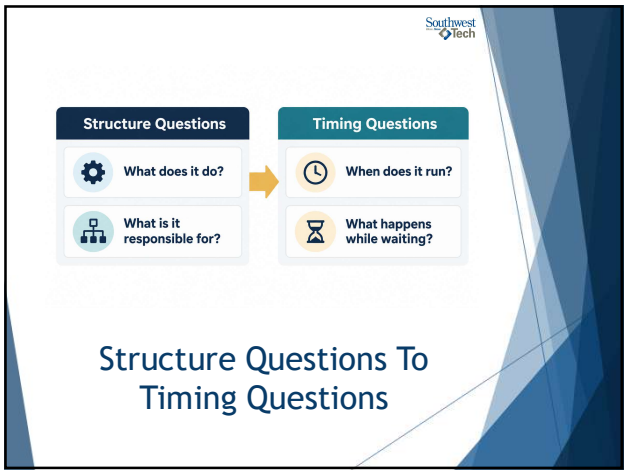
- When does this function run?
- What happens while we wait?

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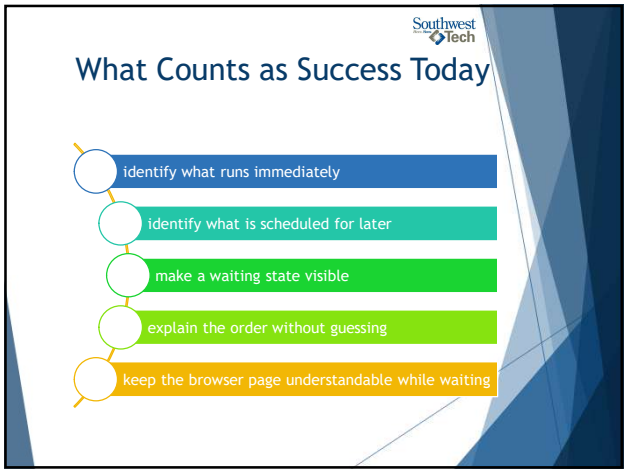


Week 7 Structure To Timing

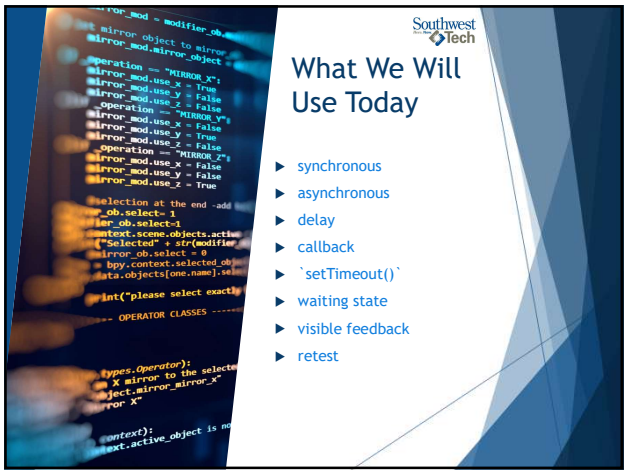
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Structure Questions
What does it do?
What is it responsible for?
Timing Questions
When does it run?
What happens while waiting?
Structure Questions To Timing Questions

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What Counts as Success Today
identify what runs immediately
identify what is scheduled for later
make a waiting state visible
explain the order without guessing
keep the browser page understandable while waiting

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What We Will Use Today
synchronous
asynchronous
delay
callback
`setTimeout()`
waiting state
visible feedback
retest

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Async Toolbox

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What We Will "Skip" For Now

Parked for later:

- full Promise chains
- advanced error handling
- API authentication
- CORS details
- service workers
- complex loading systems

For now:

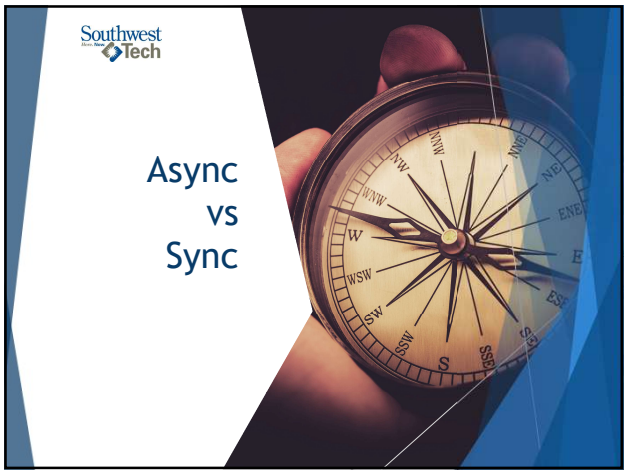
- Make timing visible.
- Explain the order.

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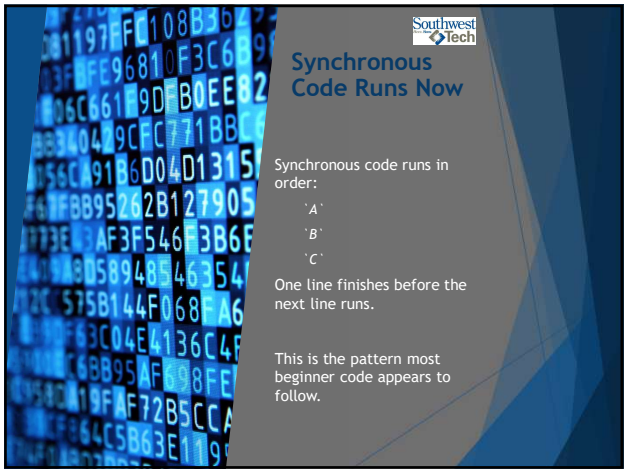


Parked For Later Async Topics

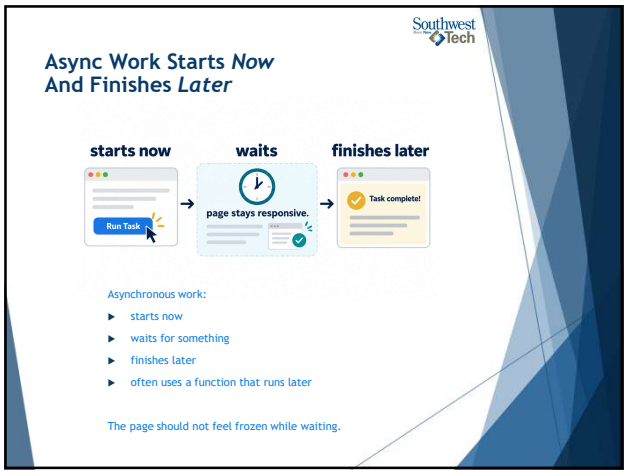
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Async
vs
Sync

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Synchronous Code Runs Now
Synchronous code runs in order:
`A`
`B`
`C`
One line finishes before the next line runs.
This is the pattern most beginner code appears to follow.

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Async Work Starts Now And Finishes Later
starts now
waits
finishes later
Run Task
page stays responsive.
Task complete!
Asynchronous work:
starts now
waits for something
finishes later
often uses a function that runs later
The page should not feel frozen while waiting.

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Python Bridge: Same Timing Question

Python and JavaScript use different syntax.

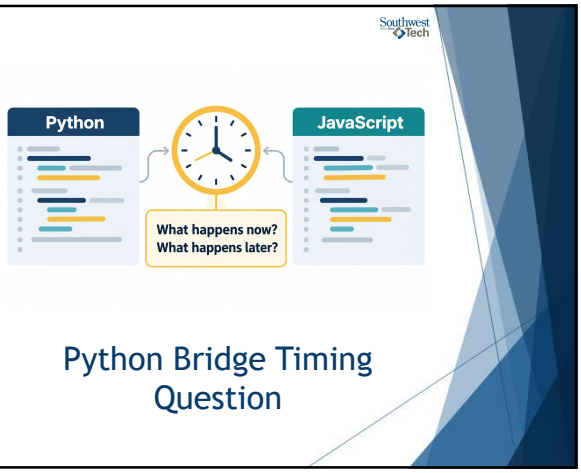
But the core question is familiar:

- What happens now?
- What happens later?
- What waits?
- What keeps going?

The timing problem is shared.



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`setTimeout()`

Makes Waiting Visible

`setTimeout()` says:

Run this function later.

It needs:

- a function to run later
- an amount of time to wait

The waiting is real, even when the code is short.



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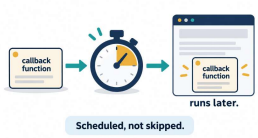
Callback: The Later Function

A callback is a function used later.

For Week 8, read it as:

"When the timer finishes, run this."

The callback is not skipped. - It is scheduled.



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User Feedback During Waiting



Good timed behavior gives feedback:

- starting
- waiting
- finished

Example:

Click button -> "Waiting..." -> "Finished."

The user should not have to guess.

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AI Can Explain Timing, Not Replace Your Work

AI as Explainer - You may ask:

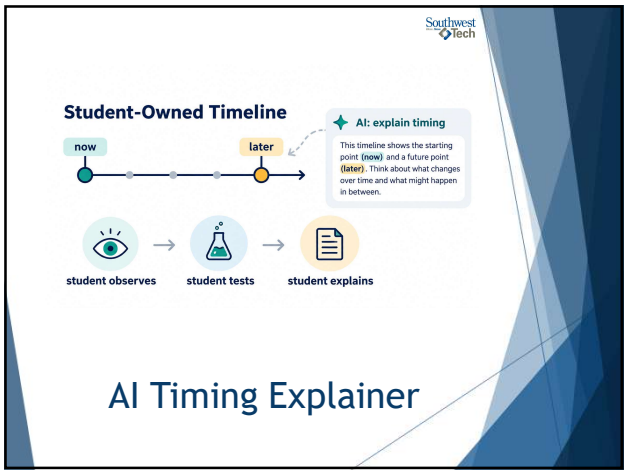
"Explain why this message appears after the other message."

You may not submit:

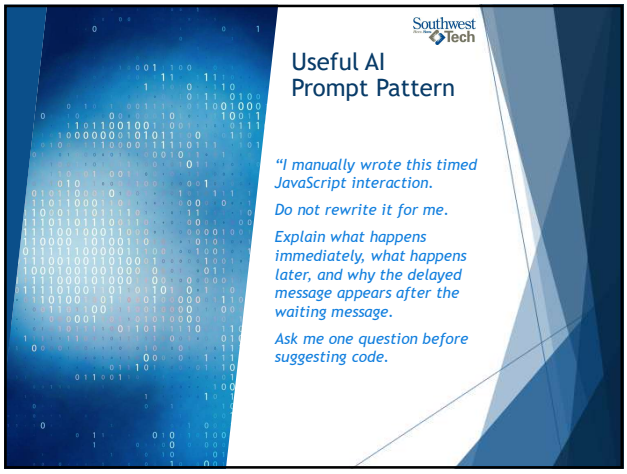
- AI-generated timed behavior you cannot explain
- code pasted without testing
- a reflection written from AI's point of view

You still observe, test, and explain.

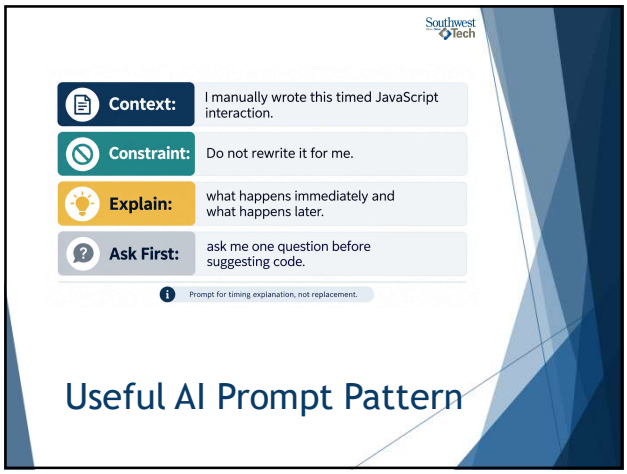
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Student-Owned Timeline
now
later
AI: explain timing
This timeline shows the starting point (now) and a future point (later). Think about what changes over time and what might happen in between.
student observes
student tests
student explains
AI Timing Explainer

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Useful AI Prompt Pattern
"I manually wrote this timed JavaScript interaction.
Do not rewrite it for me.
Explain what happens immediately, what happens later, and why the delayed message appears after the waiting message.
Ask me one question before suggesting code.

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Context:
I manually wrote this timed JavaScript interaction.
Constraint:
Do not rewrite it for me.
Explain:
what happens immediately and what happens later.
Ask First:
ask me one question before suggesting code.
Prompt for timing explanation, not replacement.
Useful AI Prompt Pattern

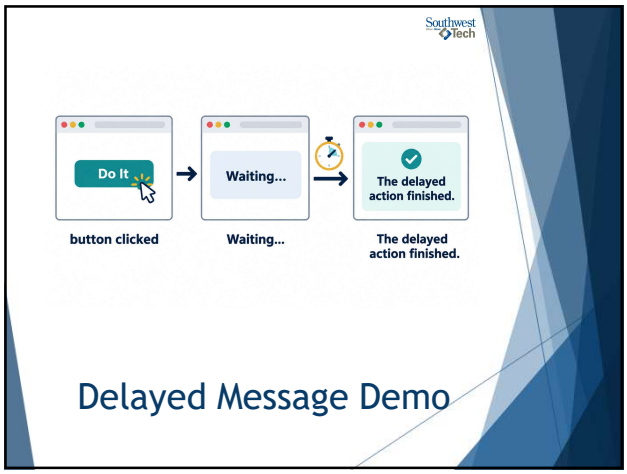
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Demo #1
"Delayed Message"

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What to Watch For
Demo:
Button click starts a delay.
The page shows:
immediate waiting message
delayed completion message
Watch for:
What happens now?
What happens later?
Q2
Q3

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Do It
Waiting...
The delayed action finished.
button clicked
Waiting...
The delayed action finished.
Delayed Message Demo

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Trace The Order



1. The user clicks.
2. The page shows "Waiting..."
3. The timer starts.
4. Other browser work can continue.
5. The delayed function runs.
6. The page shows the final message.

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Lab 08



Tuesday lab:

Build visible timed behavior.

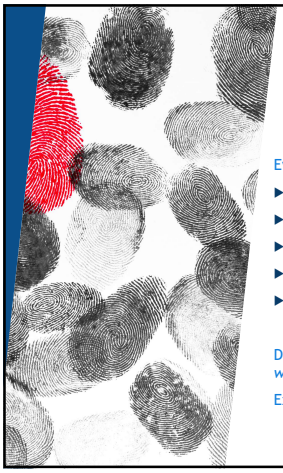
Your page should show:

- a start action
- a waiting state
- a later result

Keep the feature small enough to explain.

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Evidence



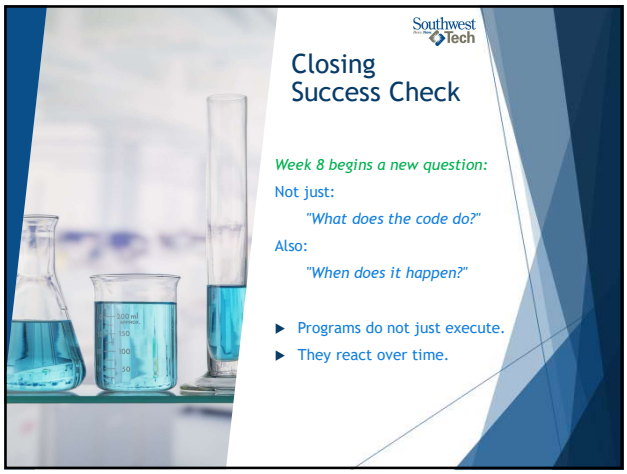
Evidence this week:

- the timed behavior is visible
- the code runs without errors
- the waiting state is clear
- the final result appears later
- your reflection explains timing

Do not rely on "*it eventually worked.*"

Explain the order.

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Closing
Success Check
Week 8 begins a new question:
Not just:
"What does the code do?"
Also:
"When does it happen?"
Programs do not just execute.
They react over time.

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Thank you!
Have Fun!

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